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REVIEW

Unsupervised machine learning methods and emerging applications in healthcare

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Abstract

Unsupervised machine learning methods are important analytical tools that can facilitate the analysis and interpretation of high-dimensional data. Unsupervised machine learning methods identify latent patterns and hidden structures in high-dimensional data and can help simplify complex datasets. This article provides an overview of key unsupervised machine learning techniques including K-means clustering, hierarchical clustering, principal component analysis, and factor analysis. With a deeper understanding of these analytical tools, unsupervised machine learning methods can be incorporated into health sciences research to identify novel risk factors, improve prevention strategies, and facilitate delivery of personalized therapies and targeted patient care.

Level of evidence: I

Keywords Machine learning · Editorial · Artificial intelligence · Computational models · Analytics

Introduction

Machine learning is an important analytical tool that can advance our understanding of human physiology and help improve patient care. In general, many research studies aim to model the relationship between a set of dependent variables and clinically relevant health outcomes. However, biological data are often complex, thereby limiting our ability to model straightforward relationships between predictors and outcomes. In these circumstances, machine learning algorithms can process clinical data and implement performance metrics to build models that elucidate complex relationships [3]. Computational models derived through machine learning can then be applied in clinical settings to improve risk stratification, facilitate early diagnosis, and design personalized treatment plans [5].

While a wide array of machine learning methods have been developed, the overarching study aims in a given study generally drive the selection of the optimal machine learning approach. Classical machine learning methods can be designated as either supervised or unsupervised methods. Supervised machine learning methods predict health outcomes based on labeled data. For example, in a recent study that aimed to identify clinical variables predictive of functional status after hip arthroscopy, a supervised machine learning method called least absolute shrinkage and selection operator (LASSO) identified key clinical variables that were predictive of post-operative functional status [14]. In this case, important predictive features were selected from labeled data, which is a defining feature of supervised machine learning methods.

In comparison, unsupervised machine learning methods identify patterns in unlabeled datasets. Specifically, unsupervised techniques process unlabeled data and can elucidate hidden structures in the data. K-means clustering

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and hierarchical clustering are two classical unsupervised machine learning techniques that partition data into groups based on similarities in the data [16]. In comparison, principal component analysis and factor analysis are unsupervised dimensionality reduction tools that can simplify and increase interpretability of high-dimensional datasets (Fact Box) [10]. In this article, we provide an overview of foundational unsupervised machine learning techniques and introduce key concepts to facilitate the application of unsupervised machine learning methods in health sciences research.

Unsupervised machine learning algorithms

K-means clustering is an iterative algorithm that partitions data points into distinct and non-overlapping clusters (Fig. 1). The unsupervised algorithm aims to minimize within-cluster variation, or the amount by which the observations in a cluster differ from each other. For example, in a recent study of bloodborne markers of joint tissue

Fact Box: Unsupervised machine learning methods

- K-means clustering is an unsupervised machine learning algorithm that partitions data into distinct and non-overlapping clusters. The algorithm requires the user to input unlabeled continuous variables and to pre-specify the optimal number of clusters.
- Hierarchical clustering is an alternative clustering approach that generates a nested clustering hierarchy that can derive multiple clustering solutions. The user must specify the measure of dissimilarity used to drive cluster formation and must determine if the clustering solution has identified meaningful structures in the data.
- Principal component analysis is a dimensionality reduction tool that transforms high-dimensional data into linear functions that explain the total variance in the data. The user must standardize and scale the inputted data and must decide how many principal components to use in subsequent analyses.
- Factor analysis is an alternative dimensionality reduction tool that extracts the common variance in the dataset in the form of unobserved or latent factors. The user must pre-specify the optimal number of factors and must ensure that there is some degree of correlation in the dataset.

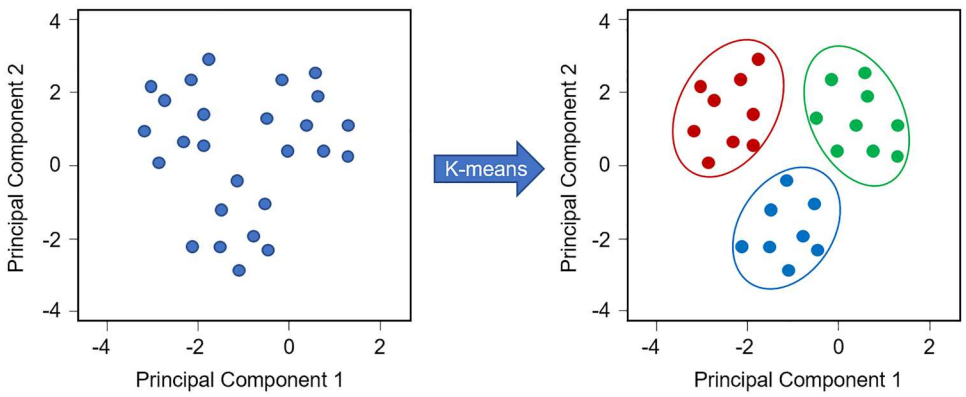


Fig. 1 Principal Components Plot of K-means Clusters. K-means clustering partitions observations into distinct and non-overlapping clusters. The algorithm partitions data points into a predetermined number of clusters (k) that minimizes within-cluster variation

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turnover, K-means clustering of biological markers was used to identify subgroups of patients with distinct osteoarthritis sub-phenotypes [2]. K-means clusters can also be included in a health model as a categorical variable in place of high-dimensional biomarker data to determine if clusters are associated with key health outcomes [7]. In general, K-means clustering represents a straightforward approach to partitioning high-dimensional data into distinct groups.

To cluster data into coherent groups, the user must specify the optimal number of clusters (k) [15]. While data-driven methods can help identify the optimal number of clusters [6], cluster selection should hinge predominantly on expert knowledge and interpretability of the results. Once the user has defined k, the algorithm randomly assigns k centroids for each cluster. The algorithm then iteratively moves each data point to the closest centroid and calculates the centroid mean for each cluster [17]. K-means then re-assigns observations to the next closest centroid, and re-iterates this process until the algorithm converges on a clustering solution that minimizes within-cluster variation. The final cluster assignments will reduce high-dimensional data into distinct and coherent groups.

When applying K-means clustering in a research environment, it is important to understand the limitations and assumptions underlying the approach. K-means clustering requires continuous variables and assumes that inputted variables comprehensively represent the underlying construct that is driving the cluster formation [12]. In general, it is favorable to include as many variables as possible that may contribute to the underlying construct (i.e., sub-phenotypes of a particular disease), knowing that variables that do not encompass substantive variance will have smaller contributions to cluster formation. The robustness of the clustering solution should then be tested in sensitivity analyses using different combinations of variables. Finally, K-means clustering is largely an exploratory method. Clustering solutions can be internally validated using bootstrapping methods that test clustering solutions in small simulated samples taken from the overall dataset [21]. Alternatively, clustering solutions can be externally validated in replication cohorts. However, the validity of the clustering solution ultimately hinges on the clinical expertise and the utility of the clustering solution.

Hierarchical clustering is an alternative clustering approach that is useful when the optimal number of clusters is unclear or the separation between data points is less delineated [9]. Unlike K-means clustering, which requires the user to establish the optimal number of clusters to start configuring the data, hierarchical clustering does not require the user to pre-specify the number of groups. Instead, hierarchical clustering requires the user to specify the measure of dissimilarity that will be used to drive cluster formation [8]. Hierarchical clustering then produces nested clusters by merging similar observations into coherent groups.

Hierarchical clustering builds clusters using an agglomerative (bottom-up) or divisive (top-down) approach [1]. The agglomerative approach begins with individual observations in the dataset. Individual observations with the smallest pairwise dissimilarity are then merged to produce a cluster. Similar clusters are then sequentially combined into larger clusters until finally there is only one remaining cluster that contains all the data (Fig. 2). In comparison, the divisive approach begins with one cluster that includes all the observations in the dataset. At each level of the hierarchy, a single cluster is split to produce two new clusters with the greatest between-group dissimilarity. The clusters are then sequentially divided until the number of clusters equates to the number of data points in the dataset.

To derive a clustering solution, hierarchical clustering requires the user to define a measure of dissimilarity between observations. Observations can be linked using single, complete, or average linkage. Each approach computes all pairwise dissimilarities between observations in a cluster,

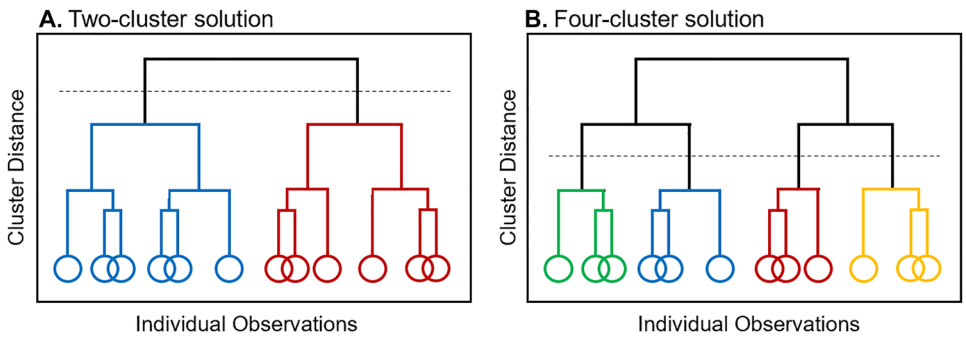


Fig. 2 Hierarchical clustering dendrograms. Agglomerative hierarchical clustering starts with individual observations. Individual observations that are similar are combined to form clusters, and similar clusters are then fused. The method produces a set of nested clusters that are arranged in the form of a tree or dendrogram. The dendrogram can then be "cut" at different levels to produce different clustering solutions including **A** a two-cluster solution or **B** a four-cluster solution

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but uses a different formula to merge clusters together. While single or complete linkage may be appropriate in some analytical scenarios, average linkage represents a compromise between single and complete linkage and tends to produce clusters that are relatively robust and compact [22].

Regardless of the measure of dissimilarity implemented in a dataset, hierarchical clustering will provide a complete description of the clustering hierarchy in the form of a graphical tree or dendrogram. The dendrogram can then be "cut" horizontally at different levels to partition the data into coherent clusters. In general, the clustering solution is valid to the extent that the hierarchical structure imposed by the algorithm reflects true similarities in the data. Thus, the user should determine if the solution has identified meaningful structures in the data. For example, in a study of bone stress injuries among collegiate runners, hierarchical clustering was used to identify groups of athletes with similar joint kinematics and kinetics [13]. Each group had distinct biomechanical metrics that translated into differential risk of bone stress injury, confirming that the clustering approach identified a meaningful structure in relation to the data. As demonstrated in this study, the user should ultimately use clinical expertise to evaluate the utility of the clustering solution and should explore alternative clustering solutions to evaluate the robustness of the results.

Clustering techniques aim to identify coherent subgroups in the data, which can elucidate unlabeled structures that carry great physiological significance. Related unsupervised machine learning techniques aim to reduce dimensionality of the data while retaining important trends in the data. Principal component analysis and factor analysis are two dimensionality reduction tools that can simplify and increase interpretability of high-dimensional datasets [10].

Principal component analysis (PCA) transforms high-dimensional data into Principal Components (PCs) that are linear functions of the original dataset. The PCs simplify high-dimensional data into fewer variables while retaining important patterns in the data [1]. PCA creates the first PC by minimizing the perpendicular distance between each observation and the linear PC, thereby maximizing the variance of the observations in the PC. Subsequent PCs are selected in the same manner, but are not correlated with the previous PCs (Fig. 3) [4]. The first PC has the greatest variance, and most of the variance will be contained in the first few PCs. The resulting PCs can then be included as the exposure of interest in health models. In general, the user must decide how many PCs to include in subsequent analyses. For example, in a study of knee joint biomechanics in knee replacement patients, PCA was applied to knee joint kinematic data [23]. The first 11 PCs accounted for 91% of the variation in the kinematic data and were retained in their analyses, which identified targetable limitations in post-operative joint mobility.

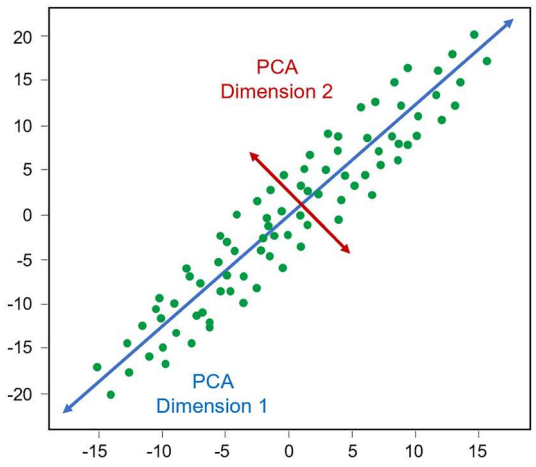


Fig. 3 Dimensionality Reduction Using Principal Component Analysis. Principal component analysis (PCA) reduces the dimensionality of data while retaining important trends and patterns. In general, the first principal component has the largest variance, and most of the variance is contained in the first few principal components

When performing a PCA, it is important to understand the strengths and limitations of the approach. First, it is essential to standardize the variables in the analysis. If the input data are measured on different scales or in different units, the PCA will be heavily weighted along high-magnitude variables and will ignore lower-magnitude variables. Input variables should therefore be centered and scaled prior to conducting a PCA [11]. Second, a PCA may miss non-linear or non-orthogonal patterns in the data. In addition, PCA may not identify clusters in the data as PCA seeks to maximize variance in linear PCs. It is therefore important to visualize the underlying data prior to conducting a PCA to determine if PCA is an appropriate approach.

While PCA seeks to explain the total variance in a given dataset, *factor analysis* is an alternative dimensionality reduction tool that elucidates common sources of variation in the data. A factor analysis extracts the common variance from a dataset and reduces a large number of variables into a smaller number of unobserved or latent factors (Fig. 4) [19]. The factors represent the common sources of variation in the data and account for their correlation structure, but do not necessarily include the variance of all variables in the dataset. Once key factors have been identified, the resulting factors can be included as the exposure of interest in health models. For example, in a study evaluating a 15-item patient-reported outcome tool among patients with chronic ankle instability, a factor analysis identified two key factors that were associated with psychological deficits and decreased physical activity among study participants [20]. A follow-up analysis could then evaluate whether these two factors are associated with longitudinal health outcomes

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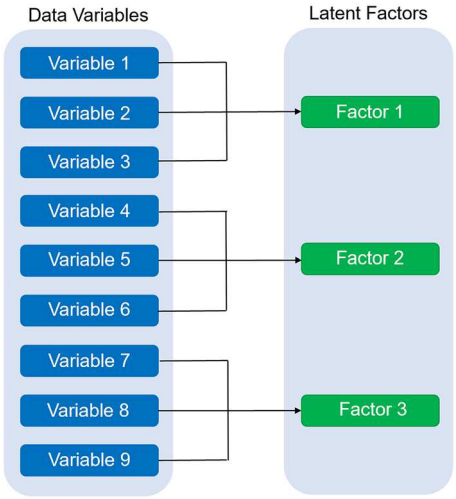


Fig. 4 Dimensionality reduction using factor analysis. Factor analysis reduces dimensionality of data by identifying common sources of variation otherwise known as latent factors

such as the possible need for surgery among patients with chronic ankle instability.

When implementing a factor analysis, it is important to understand key steps and computational assumptions. Unlike PCA, factor analysis requires the user to pre-specify the optimal number of factors. In general, clinical expertise and interpretability should guide factor selection. However, factor loadings can help with factor selection by indicating the degree of correlation between factors and the underlying variables [18]. Stronger correlation between the factor and the underlying variables confirms that the factor extracts a significant amount of variance from the data. In addition, factor analysis assumes there is some degree of correlation in the data, and that a weighted composite of latent factors can encompass the values of the variables in the dataset. In general, factor analysis is useful when there are underlying relationships or correlations between the variables in a dataset that can be captured by latent factors.

Conclusion

Unsupervised machine learning methods including clustering techniques and dimensionality reduction tools carry great potential to elucidate structures and patterns in high-dimensional data and expand understanding of human physiology and disease. In general, the overarching research aims in a study should drive the selection of the appropriate machine learning tools, and a clear understanding of the assumptions and limitations of machine learning methods is required. With these considerations in mind, unsupervised machine learning techniques can be incorporated into health sciences research to identify novel risk factors, improve prevention strategies, and facilitate delivery of personalized therapies and health care.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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